

FYP - OpenPaths

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Contents

1	Introduction	3
2	Literature Review	3
2.1	Fundamental Concepts in Road Network Modelling	4
2.1.1	Graph Representation of Transport Networks	4
2.1.2	Flow and Demand Representation	4
2.1.3	Network Evaluation Frameworks	5
2.2	Optimisation Approaches for Network Design	7
2.2.1	Heuristic Approaches	7
2.2.2	Linear Programming and Mixed Integer Linear Programming	8
2.2.3	Metaheuristic Approaches	9
2.3	Conclusion	9

List of Acronyms

ABM Agent-Based Modelling.

AT-ABM Active Transportation Agent-Based Model.

BNDP Bicycle Network Design Problem.

C-SAM Connections for Safer Active Mobility.

CC Closest Component.

DNDP Discrete Network Design Problem.

L2C Largest-to-Closest.

L2S Largest-to-Second.

LP Linear Programming.

MILP Mixed Integer Linear Programming.

MPAM Multi-Class Pedestrian Assignment Model.

NHTS National Household Travel Survey.

NSGA-II Non-Dominated Sorting Genetic Algorithm II.

NSO National Statistics Office.

OD Origin–Destination.

R2C Random to Closest.

SPSA Simultaneous Perturbation Stochastic Approximation.

TOPSIS Technique for Order Preference by Similarity to Ideal Solution.

UNDP Urban Network Design Problem.

1 Introduction

Transportation systems influence multiple sectors of society, shaping the environment, public health, and socioeconomic relationships. Decades of car-centric urban development have contributed to rising carbon emissions, escalating road maintenance costs, and the creation of hostile public spaces that discourage active mobility.

A substantial portion of national expenditure is directed toward road construction and maintenance. According to CEIC data [1], Malta spent a median of approximately €21 million annually between 1995 and 2014. Despite these investments, road safety remains a significant issue. National Statistics Office (NSO) [2] reported around 16,000 road accidents in 2023, including 26 incidents involving cyclists, eight of whom sustained grievous injuries. Such figures highlight both the financial and societal costs of car dependency for the country as well as the inherent dangers to cyclists.

Maltese citizens do not see the country as being apt to cycle in, as reflected in the National Household Travel Survey (NHTS) of 2021 [3], which reported that only 0.7% of all trips were carried out by bicycle. Respondents identified road safety concerns (17%) and poor-quality cycling infrastructure (13%) as the principal barriers to cycling. The Maltese government has shown commitment to improve the local cycling infrastructure in the recent years, with the ongoing project "Vjal Kulhadd" launched in 2024 [4], with an investment of 10 million euros [5]. A segregated passageway dedicated to pedestrians and cyclists from Zabbar to Żejtun was completed in March 2025 [6], with the first phase of this project planned to be completed by 2026. Another project is the Connections for Safer Active Mobility (C-SAM) initiative, which is a nationwide cycling and active-mobility network plan for Malta announced in 2022, with an estimated budget of €35 million and a target of 50-60 km of new cycling routes [7]. The Grand Harbour Cycling Network is the first component of C-SAM with a budget of €27.4 million, focusing on the Grand Harbour area (encompassing Msida, Pietà, Blata l-Bajda, Valletta, Marsa and Paola) and is scheduled for completion in phases by 2028 [8].

In parallel, government initiatives are also targeting public transport improvements to reduce reliance on private vehicles. These include the addition of 25 new buses in 2025 [9] and renewed discussions around the proposed Malta Metro system, initially announced in 2021 [10], temporarily shelved, and later revived for consultation in 2025 [11].

Together, these efforts reflect a growing recognition of the need for a balanced and sustainable mobility system, where active and public transport play central roles in shaping Malta's urban future. Transitioning cities toward walkable and cyclable environments can improve public health, reduce emissions, and foster safer and more inclusive urban spaces. To support this transition to a more expansive cycling network, there is a growing need for data-driven tools that assist urban planners in reallocating road space from private cars to active mobility users.

The overarching aim of this project is therefore to develop a computational framework that optimally expands Malta's cycling network, ensuring safe and continuous connections between major destinations while maintaining a functional and connected road system for cars. The current transport context in Malta underscores the urgency of this objective.

2 Literature Review

Designing efficient and safe bicycle networks requires an understanding of how transport systems can be represented, analysed, and optimised using quantitative models. The literature on network design spans a range of approaches that variously emphasise network structure, infrastructure optimisation, and traveller behaviour. Reflecting this diversity, this review first outlines the fundamental concepts common to all modelling frameworks, including graph representation, flow estimation, and network evaluation. Then examining the main optimisation approaches used in bicycle network design: heuristic, linear programming and mixed-integer linear programming, and metaheuristic

2.1 Fundamental Concepts in Road Network Modelling

2.1.1 Graph Representation of Transport Networks

All network design frameworks represent the underlying road infrastructure as a graph, where intersections correspond to nodes and road segments to edges. The network is typically modelled as a directed multigraph, meaning that edges may carry flow in one or both directions, and multiple edges can exist between the same pair of nodes to represent different lanes or road types, an example of such a graph can be seen in fig. 1. Each edge is assigned a weight that quantifies its travel impedance, commonly travel time, which can be derived from distance and speed limit for cars. For bicycle networks, additional attributes such as perceived safety and gradient are frequently incorporated into edge weights to reflect cyclist preferences [12].

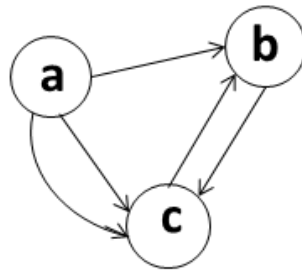


Figure 1: Example of a directed multigraph, source:[13]

An important extension of this graph representation is the use of multilayer networks, which capture the interactions between different modes of transport. In such models, each layer corresponds to a distinct mode, such as motorised traffic, cycling, or walking, while shared intersections or transfer points act as inter-layer connections. This representation allows planners to account for modal dependencies and conflicts, for example when cycling lanes share space with car traffic or when cyclists use protected corridors separated from the vehicular network. Lonardi et al. [14] explicitly adopt this formulation in their *Cohesive Urban Bicycle Infrastructure Design* framework, modelling the urban transport system as coupled road and protected-cycle layers to evaluate how improvements in one layer affect the other. Even studies that focus solely on the cycling network implicitly rely on this concept, as proposed bicycle links are embedded within an existing multilayered urban fabric that includes vehicle and pedestrian flows.

Such approaches are examples of the broader [Urban Network Design Problem \(UNDP\)](#), and more specifically the [Discrete Network Design Problem \(DNDP\)](#), which is generally considered NP-hard [12]. This complexity arises because network design involves bi-level decision-making. At the upper level, discrete infrastructure choices are made, such as reallocating or upgrading road segments. At the lower level, user flows then adjust to the new network conditions. The interaction between these two levels produces a non-convex and combinatorial search space, making it computationally intractable to find the global optimum for large-scale networks. The way travel demand and flow are represented at this lower level directly affects both the realism of user responses and the computational tractability of the overall optimisation. Accurately and efficiently capturing this interaction is therefore essential for evaluating how proposed network changes influence travel behaviour and system performance.

2.1.2 Flow and Demand Representation

Representing travel demand and flow accurately is a critical element of transport network modelling. In essence, while the graph defines the spatial structure of the transport system, flow representation determines how travellers use that structure, that is, how many trips occur between different locations and how these trips distribute across available routes. There are two main models capturing this relationship at varying levels of detail, [Origin–Destination \(OD\) matrices](#) and trajectory data.

An [OD matrix](#) defines the total number of trips between each origin and destination pair within the study area, usually derived from household travel surveys or census statistics. [OD data](#) provides a macroscopic view of mobility patterns, making it suitable for equilibrium-based and aggregate network models where flow assignment depends on total demand rather than individual behaviour. For instance, [15, 16] both rely on [OD data](#) to simulate pedestrian

and cycling flows at a city scale. Another instance, in the Tehran case study, [15] estimated the OD matrix using a combination of household surveys and field observations, such as counting pedestrians at subway stations and bus stops. Trips were classified into home-based and non-home-based categories, further subdivided into work and leisure purposes, allowing the model to distinguish between trip types with different sensitivities to distance and environmental factors. In contrast [17] assume uniform OD demand between all node pairs to isolate the topological effects of network structure.

On the other hand, trajectory data such as GPS traces, or crowdsourced cycling data (e.g., Strava), captures individual-level travel behaviour with high spatial and temporal precision. This data enables the reconstruction of actual routes taken by users, revealing preferences for safety, directness, or comfort that OD matrices cannot capture. For example, [18] employ bike trajectory data to identify observed cyclist movement patterns and use these trajectories to guide the placement of new cycling links, thereby improving the behavioural realism of their model. Trajectory datasets are therefore particularly valuable for agent-based models, where individual route choices and responses to infrastructure are explicitly modelled.

Each data source presents trade-offs. Trajectory data provides detailed behavioural information but often suffers from sampling bias, such as overrepresentation of fitness-oriented or app-using cyclists and privacy limitations, while OD matrices are more systematically collected but lack route-level detail and temporal variation.

Once travel demand and flows are defined through OD matrices or trajectory data, the next step in evaluating bicycle networks is to quantify their structural and functional performance using a range of network metrics or agent based modelling.

2.1.3 Network Evaluation Frameworks

Topological Metrics Network metrics provide quantitative indicators of how effectively a transport network supports mobility, accessibility, and resilience. These metrics are derived from graph theory and in literature are used both to evaluate existing networks and to guide heuristic or optimization models toward more efficient configurations.

Connectedness is among the most fundamental measures, describing whether the network forms a single, navigable system or remains fragmented into multiple components. Metrics such as the size of the largest connected component or the number of disconnected subgraphs are commonly used to assess overall network cohesion. High connectedness indicates that users can reach most destinations without detouring the safe cycle network, an essential property for promoting cycling continuity and accessibility.

Directness measures how efficiently users can travel between two points and is typically defined as the ratio of Euclidean distance to shortest-path distance within the network. Natera et al. [18] take a different approach to directness, calculating it as the ratio of distance travelling with the car compared to the distance travelling with the bicycle. Values closer to one indicate straighter, more direct routes and fewer unnecessary detours, important for both user comfort and competitiveness with motorised travel. Studies such as [17] use directness to evaluate how heuristic growth strategies improve route efficiency as new segments are added.

Centrality metrics capture the relative importance of nodes or links within the network. Betweenness centrality, for example, measures how frequently a link lies on the shortest paths between all node pairs, serving as a proxy for potential flow or load. High-betweenness edges often correspond to corridors that would carry the most traffic if demand were evenly distributed, making this measure a key input in centrality-based heuristics [17]. Other centrality measures, such as closeness (proximity of a node to all others), and degree (number of connections per node), indicate network accessibility and local connectivity density.

Beyond these classical indicators, Szell et al. [17] employ higher-order measures such as global and local efficiency, spatial clustering, and anisotropy to capture the balance between resilience, compactness, and directional bias. For instance, global efficiency measures the overall accessibility of a network by quantifying how efficiently any two nodes can reach each other through the existing connections, while local efficiency evaluates how well a neighbourhood remains connected if a node fails. Spatial clustering and anisotropy help identify concentration patterns and directional preferences, offering insights into the morphological coherence of urban cycling networks. Unlike purely geometric measures, perceived safety captures how infrastructure design, such as physical separation, traffic volume, or intersection treatment, affects the attractiveness and usability of cycling routes, providing a human-centred dimension to network evaluation.

Together, these metrics provide a quantitative framework for assessing and comparing networks across cities or growth strategies. They are particularly useful for heuristic and optimization approaches that aim to improve network structure without modelling detailed individual behaviour. However, as the next section discusses, agent-based models complement these static indicators by capturing the dynamic, individual-level interactions that give rise to aggregate network flows.

Agent-Based Modelling [Agent-Based Modelling \(ABM\)](#) provides a behaviourally rich framework for analysing active transportation systems. By simulating interactions between individual travellers (agents) and their surrounding urban environment, [ABMs](#) operate at the microscopic level, allowing system-wide travel patterns to emerge from the collective behaviour of heterogeneous individuals. Unlike aggregate or topological models, this bottom-up structure explicitly represents decision-making processes such as mode and route choice, enabling researchers to examine how changes in cycling and walking infrastructure influence safety outcomes, travel behaviour, and equity of access. The ability of ABMs to incorporate behavioural adaptation and spatial heterogeneity makes them particularly suited for studying the dynamic relationships between users and the built environment. However, the reliability of their outputs depends on how accurately these models reproduce real-world conditions, which in turn hinges on data quality, calibration methods, and behavioural realism.

Agent-based evaluation methods vary widely in their spatial scale, behavioural detail, and underlying data sources. Aziz et al. [19] developed the [Active Transportation Agent-Based Model \(AT-ABM\)](#) to simulate neighbourhood-scale mode choice between home and work trips. In this model, the agents' baseline probabilities of selecting a transport mode depend on attributes such as income, age, and perceived safety, and are further influenced by social interactions with other agents within the same local environment. The model integrates traffic safety records, capturing pedestrian and cyclist crash frequencies with population and socio-demographic datasets to parameterise agent characteristics. Because the [AT-ABM](#) is non-deterministic and lacks analytical gradients, it employs the [Simultaneous Perturbation Stochastic Approximation \(SPSA\)](#) algorithm for heuristic calibration, iteratively perturbing model parameters and evaluating outcome changes to efficiently estimate behavioural sensitivities under stochastic variation.

In contrast, [20] adopt a rule-based [ABM](#) to explore how the introduction of separated cycling infrastructure affects driver behaviour and road safety. Using a network of approximately 13,000 road segments, the model represents behavioural adaptation in which drivers gradually learn to anticipate cyclists after repeated encounters, leading to reduced collision risks up to a saturation point. Road types are differentiated by exposure and visibility levels, and potential crash events are simulated at intersections based on whether drivers have recently encountered cyclists. Rather than relying on statistical calibration, parameters such as driver learning rate and perception sensitivity are systematically varied across scenarios to examine emergent safety dynamics. This approach provides a dynamic behavioural perspective absent from traditional static safety models, illustrating how [ABMs](#) can capture the temporal evolution of driver awareness and risk perception.

A broader, city-scale application is demonstrated by Jafari et al. [21], who employ an integrated agent-and-activity-based simulation framework to assess changes in travel mode share resulting from proposed cycling infrastructure in Melbourne, Australia. The model makes use of census-based demographic data, [OD](#) matrices, and household travel surveys to define behavioural clusters of riders characterised by trip length, safety preference, and socio-economic background. These clusters inform the agent population, allowing simulation of how a proposed cycle corridor would alter travel behaviour across demographic groups. Calibration is achieved deterministically by iteratively adjusting mode-choice constants until simulated mode shares align with observed data. The results show that travel time strongly influences mode choice across all clusters, and that improved cycling infrastructure and perceived safety substantially increase cycling uptake, by approximately 30%, highlighting the value of agent-based approaches for infrastructure impact assessment and policy evaluation.

Collectively, these studies underscore the value of [ABMs](#) in capturing behavioural, temporal, and spatial complexity within active transportation systems. By modelling the decision-making processes of individual travellers, [ABMs](#) allow researchers to explore how and why infrastructure changes influence user choices, safety outcomes, and modal shifts, moving beyond purely structural or geometric network assessments. Their flexibility makes them particularly powerful for simulating emergent phenomena such as behavioural adaptation, social influence, and mode-shift dynamics under different policy or design scenarios.

Nevertheless, several methodological challenges remain. [ABMs](#) are data-intensive, requiring detailed demographic, safety, and travel diary information to parameterise and validate behavioural rules. Their complexity introduces large numbers of parameters, making calibration and validation difficult. Models such as that of Aziz et al. [19] address

this through stochastic optimisation techniques like [SPSA](#), while others, such as of Thompson et al. [20], rely on heuristic or manual parameter variation, potentially affecting reproducibility and generalisability. Moreover, many [ABMs](#) focus on mode choice and aggregate travel flows rather than explicit route-level movement, which may limit their geometric fidelity in representing real-world network performance. Finally, simplified behavioural assumptions, though necessary for computational feasibility, can constrain realism when modelling diverse urban contexts.

These evaluation frameworks provide the means to assess and simulate how effectively a network supports mobility and behavioural adaptation. However, evaluation alone does not determine which infrastructure changes should be implemented. The next challenge is therefore optimisation, identifying from the many possible interventions, the set of new segments or reallocations that most improve overall network performance. The following section reviews the main optimisation approaches proposed for this purpose.

2.2 Optimisation Approaches for Network Design

In the literature, the upper-level optimization problem in bicycle network design is typically addressed through one of three main approaches: heuristic, linear or mixed-integer programming, and metaheuristic methods. Each approach presents distinct advantages, limitations, and suitable use cases, balancing between computational efficiency, behavioural realism, and scalability. With this foundation on how these problems are conceptually structured, the following sections discuss the different modelling approaches in greater detail.

2.2.1 Heuristic Approaches

The simplest of the methods used in bicycle network planning are heuristic approaches. These are considered “simple” because they rely on rule-based or greedy decision-making guided by intuitive network metrics such as connectedness, directness, or betweenness centrality. Heuristic models iteratively add or remove edges in the bicycle network to improve a selected topological metric, without formulating the problem as a formal bi-level optimization task. In doing so, they make several simplifying assumptions, most notably, that traffic demand is uniformly distributed across the network and that travellers do not adjust their routes in response to infrastructure changes (i.e., no user equilibrium is modelled). As a result, these methods capture the geometric and structural characteristics of a network but neglect behavioural feedback and flow dynamics.

For instance, Sezell et al. [17] adopt a centrality-based heuristic, where new links are added according to graph-theoretic importance measures, such as betweenness or closeness centrality. This approach prioritizes edges that are topologically critical or spatially central within the underlying street network, allowing the simulated bicycle network to grow along corridors that would most efficiently shorten travel distances.

In contrast, Natera et al. [18] propose component-based heuristics designed to reconnect fragmented pieces of existing infrastructure. Their [Largest-to-Second \(L2S\)](#) strategy links the two largest disconnected components to quickly consolidate the network, while the [Largest-to-Closest \(L2C\)](#) strategy connects the largest component to its nearest neighbour, favouring shorter and more spatially efficient connections. Both approaches share the same principle of incremental, topology-driven expansion, but differ in how they define the “importance” of the next edge to add: [L2S](#) biases towards overall connectedness, whereas [L2C](#) favours directness by preferring shorter spatial gaps. The authors also use simpler baselines for comparison, using [Random to Closest \(R2C\)](#), representing uncoordinated development, and [Closest Component \(CC\)](#), which connects the nearest two subgraphs, producing a minimum spanning tree that minimises investment but is not resilient to any interruptions. Overall, [L2C](#) performed best, with [L2S](#) also strong but slightly less effective, both outperforming the baselines. The study shows that targeted investments into key missing links can rapidly consolidate fragmented bicycle networks, though cars still achieve greater directness, with routes averaging about 33% shorter than those of bicycles.

A third paradigm is introduced by Steinacker et al. [22] applies percolation-based heuristics to model network growth and robustness over time. In a forward percolation approach, the network gradually expands by adding segments with the highest marginal contribution to global accessibility or welfare. Conversely, the inverse percolation approach begins from a fully built network and iteratively removes the least critical segments, recalculating connectivity and demand after each step. An additional advantage of the inverse percolation method is that it inherently maintains full network connectivity throughout the process, as segments are only removed if their deletion does not fragment the network. This ensures that all resulting configurations remain fully functional while progressively identifying the

most critical corridors. These methods provide a transparent way to approximate how incremental investments or disinvestments affect overall network integrity.

Heuristic models consistently outperform baseline random growth strategies. Centrality-based growth yields higher connectedness, directness, and resilience than both random and minimum spanning tree baselines, producing networks that more closely resemble those of well-developed cycling cities such as Copenhagen [17]. Component-based heuristics (L2S and L2C) outperform random or closest-component baselines by achieving faster consolidation and greater overall cohesion [22]. These findings demonstrate that, even under uniform demand assumptions, simple topology-driven heuristics can reproduce high-quality, cohesive bicycle networks far more efficiently than unguided or piecemeal local planning.

While all heuristic families improve structural aspects of the network, they share fundamental simplifying assumptions. All assume uniform travel demand across the city and neglect behavioural factors such as route choice, mode preference, or induced demand. Centrality-based growth interprets betweenness as a proxy for flow under uniform conditions, component-based heuristics connect infrastructure purely by geometry, and percolation-based methods evaluate robustness statically without modelling user adaptation. Consequently, heuristic approaches provide a computationally efficient and transparent means of exploring large-scale bicycle networks and identifying missing or critical links, but their simplicity comes at the cost of behavioural realism, dynamic feedback, and global optimality.

2.2.2 Linear Programming and Mixed Integer Linear Programming

Linear Programming (LP) and Mixed Integer Linear Programming (MILP) formulations represent the most analytically rigorous class of methods for solving the Bicycle Network Design Problem (BNDP). LP models optimise continuous allocation decisions, such as how much existing road capacity should be assigned to cyclists versus cars, while MILP extends this framework by introducing integer or binary variables to represent discrete choices such as whether a new link should be constructed or upgraded. This enables more realistic representation of infrastructure planning decisions, but substantially increases computational complexity. Both LP and MILP approaches treat network planning as a constrained optimization problem, balancing physical space limits, budgetary constraints, and multimodal interactions to identify globally optimal infrastructure configurations. Compared to heuristic methods, these approaches explicitly model the trade-offs between conflicting objectives, such as cycling accessibility, car travel time, and construction costs, within a mathematically defined feasible space.

While these models guarantee mathematical optimality and offer interpretable policy trade-offs, their precision comes at a computational cost. Solving large-scale MILPs with spatial or behavioural constraints remains NP-hard, often requiring simplifications such as relaxations (temporarily allowing integer variables to take continuous values), decomposition (solving smaller subproblems sequentially), or iterative rounding (progressively fixing variables) to maintain tractability [12]. These simplifications improve computational efficiency but sacrifice global optimality, resulting in solutions that approximate the best network configuration rather than guaranteeing it. Moreover, MILP based formulations generally rely on simplified behavioural assumptions, for example, fixed OD demand, where the total number of trips between origins and destinations remains constant, and static route choice, where users follow a single optimal path that does not adjust to changing network conditions. These assumptions make MILP frameworks computationally efficient but limit their ability to capture behavioural feedbacks such as induced demand, route adaptation, or mode shifts, which are better represented in agent-based models.

Wiedemann et al. [12] propose a multi-criteria linear programming model that allocates limited street capacity between cars and cyclists. The model minimises the combined travel time of both modes, weighted by a tunable parameter that expresses the policy preference between car and bicycle performance. To reduce computational cost, the authors apply a linear relaxation of the integer lane-allocation problem and iteratively fix the highest-capacity edges after each run, progressively reducing the search space. The final model is able to handle 830-edge networks on a single core up to 10,000 edge networks using parallel processing. The approach produced a 35 % improvement in cycling performance with less than a 20 % increase in car travel time, outperforming heuristic baselines. However, the model tends to allocate one lane of each two-lane road to cyclists, neglecting detailed congestion dynamics for cars.

Steinacker et al. [22] extend this formulation through a temporal MILP framework that optimizes the expansion of Copenhagen's cycle superhighway network under budget and scheduling constraints. By incorporating time-discounted benefits, their model evaluates not only where but also when infrastructure should be implemented, supporting phased investment planning. Similarly, Lonardi et al [14], improved congestion modelling resulting in decreased car usage and increased cycling usage over time. Together, these works demonstrate how temporal and multimodal extensions of

MILP formulations enhance the policy relevance of optimization-based network planning. Although [22] found that MILP achieved a marginally higher welfare metric outcome, the key differences lay in the temporal order and specific corridors selected for construction, not in the total level of network performance. Their analysis showed that demand uncertainty has minimal effect on results as it averages across many OD pairs, whereas cost uncertainty significantly affects network performance, particularly in direct optimization models.

LP and MILP methods provide a mathematically rigorous foundation for bicycle network design, enabling planners to quantify explicit trade-offs between modes, budgets, and temporal phases. They deliver transparent and reproducible optimization outcomes and often outperform heuristic approaches in precision and efficiency. However, their high computational complexity and reliance on simplified behavioural assumptions such as fixed demand and static route choice limit their scalability and behavioural realism. To address these challenges, research has explored metaheuristic optimization that can handle multi-objective problems more flexibly.

2.2.3 Metaheuristic Approaches

Metaheuristic approaches aim to balance realism and computational feasibility, modelling the real world more accurately than simple heuristics while avoiding the exhaustive search required by exact MILP formulations. To achieve this, [15, 16] employ the non-dominated Sorting **Non-Dominated Sorting Genetic Algorithm II (NSGA-II)** to solve multi-objective optimization problems that simultaneously consider conflicting objectives for cyclists and car users. NSGA-II is an evolutionary algorithm that iteratively evolves a population of candidate solutions, ranking them based on their Pareto dominance i.e. how well they perform relative to others across multiple objectives without being strictly better or worse in all of them. The outcome is a Pareto front, a set of optimal trade-off solutions where improving one objective (e.g., cyclist safety or travel time) would necessarily worsen another (e.g., car travel efficiency). Within this front, crowding distance is used to preserve diversity among non-dominated solutions, preventing premature convergence to a single region of the search space. After convergence, the final non-dominated set is further refined using the **Technique for Order Preference by Similarity to Ideal Solution (TOPSIS)** method to rank and select the most balanced designs. This approach enables planners to explore a spectrum of balanced network configurations rather than a single, rigid optimum.

In Rastbin et al. [15] when evaluating the model on the large-scale case study on the historical-Cultural district of Tehran, Iran (approximately 2,000 hectares, 55 nodes, and 98 edges), the proposed NSGA-II bi-level model demonstrated strong computational efficiency and scalability. The lower-level **Multi-Class Pedestrian Assignment Model (MPAM)** was solved with an accuracy threshold of 10^3 , ensuring convergence while maintaining tractability for the large network. The optimization process required approximately 8,100 MPAM evaluations, compared to over 1,048,000 evaluations that would be needed under full enumeration, representing a significant reduction in computational effort. The test network included roughly 1,800 OD pairs representing both work and leisure trips. The resulting Pareto front provided planners with a spectrum of trade-off solutions balancing travel cost, environmental quality, and construction cost, confirming that the NSGA-II approach can efficiently solve complex, behaviourally informed network design problems at a practical scale.

2.3 Conclusion

The reviewed studies demonstrate that bicycle network design can be approached through a range of methodological frameworks. Together, these methodological perspectives reveal a landscape of complementary tools rather than a single dominant paradigm. However, a persistent challenge lies in integrating behavioural realism and computational scalability within a unified optimisation–evaluation framework. Addressing this gap motivates the present work, which aims to develop a modelling tool that combines scalable network optimisation with behaviourally informed evaluation to support evidence-based planning of cycling infrastructure in Malta.

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